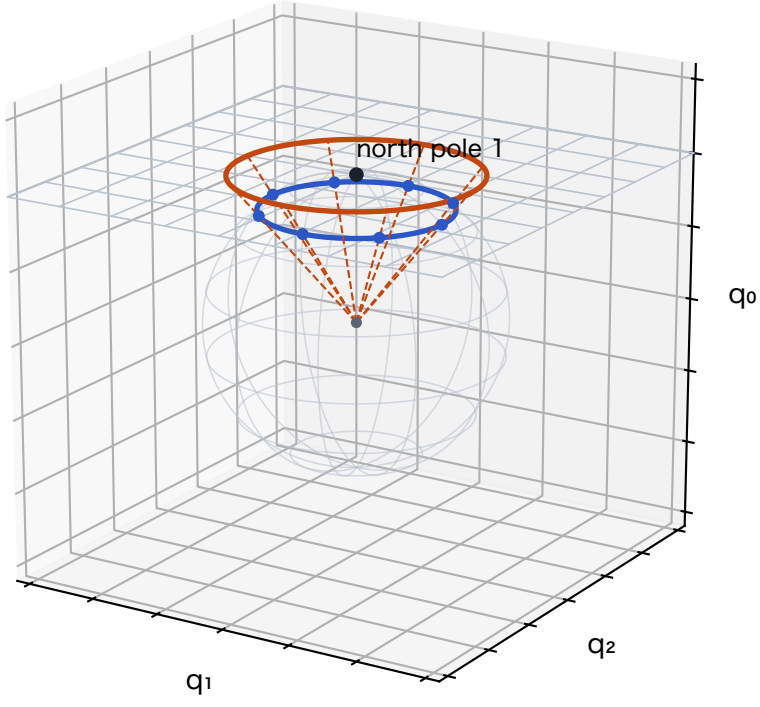
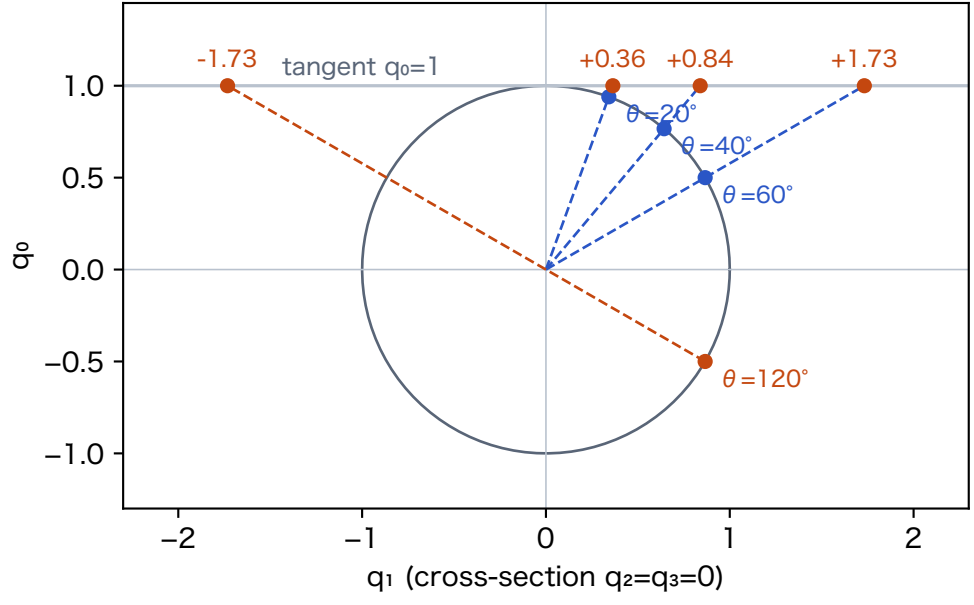
Figure 3. Central projection onto the tangent plane at the north pole (gnomonic projection): $u = v/q_0 = n \tan\theta$

3D slice $q_3=0$ (q_0, q_1, q_2): latitude circle at $\theta=40^\circ$ (blue) and its image on the tangent plane $q_0=1$ (orange, radius $\tan\theta$)



Cross-section: a point at angle θ maps to $\tan\theta$ on the tangent. The point at $\theta=120^\circ$ passes through the origin and maps to the same point as $-q$ ($\tan 120^\circ = -1.73$)